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$$V = \pi\int_0^4 r^2 dx$$

$$= \pi\int_0^4 \left[\left(2-\sqrt{x}\right)^2\right] dx$$

By Shell Method.

$$V = 2\pi\int rpw$$

$$= 2\pi\int_0^2 (2-y)\left(y^2-0\right) dy$$

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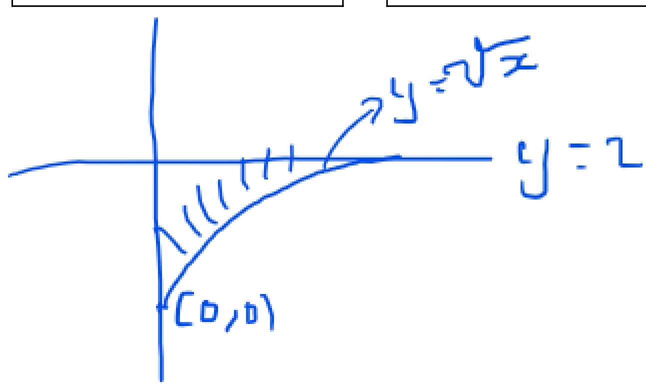

U-substitiution.

$$\int \underbrace{f}_{\text{Outside function}}\Bigg(\overbrace{g}^{\text{inside function}}(x)\Bigg)\overbrace{g'(x)}^{\text{Derivative of inside function}}\,dx = F(g(x)) + C$$

Recognizing the $f(g(x))g'(x)$ form.

$$\int x\left(x^2+1\right)^2 dx$$
$$u = x^2+1$$
$$du = 2xdx$$
$$\frac{dy}{2} = \frac{2xdx}{2} \Rightarrow xdx = \frac{du}{2}$$
$$\int x\left(x^2+1\right)^2 dx \Rightarrow \int \frac{1}{2}\cdot(u)^2\,du$$
$$= \frac{1}{2}\int u^2 du$$
$$= \frac{1}{2}\cdot\frac{u^3}{3} + C$$
$$\frac{u^3}{6} + C$$

$$\int \sqrt{2x-1}dx$$
$$u = 2x-1 \Rightarrow du = 2dx$$
$$\int \sqrt{u}\cdot\frac{du}{2}$$
$$\frac{1}{2}\int u^{1/2}du$$
$$\frac{1}{2}\cdot\frac{2}{3}u^{3/2} + C$$
$$\therefore \int \sqrt{2x-1}dx = \frac{1}{3}\cdot(2x-1)^{3/2} + C$$

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Change of Variables.

$$\int_0^1 x\left(x^2+1\right)^3 dx$$

Let $u=x^2+1$

$$du = 2xdx$$

$$\frac{1}{2}du = xdx$$

$$\boxed{\begin{array}{ll} x=0 & x=1 \\ u=1 & u=2 \end{array}}$$

$$\int_0^1 x\left(x^2+1\right)^3 dx \Rightarrow \frac{1}{2}\int_1^2 u^3 du$$

$$=\frac{1}{2}\cdot\frac{u^4}{4}\Big]_1^2=\frac{15}{8}$$

Integration of Trigonometric Functions.

$$\int \sec x dx$$

$$\int \frac{\sec x\left(\sec x+\tan x\right)}{\left(\sec x+\tan x\right)}dx$$

$$\int \frac{\sec^2 x+\sec x\tan x}{\sec x+\tan x}dx$$

$$\ln\left|\sec x+\tan x\right|+C$$

1

$$\int\left(1+\cot^2 x\right)dx$$

$$\int \csc^2 x dx = \cot x + C$$

#2

$$\int \frac{(\ln x)^3}{x}dx \qquad u = \ln x \qquad du = \frac{1}{x}dx$$

$$\int u^3 du$$

$$\frac{1}{4}u^4 + C$$

$$\frac{1}{4}(\ln x)^4 + C$$

#10 Diagnostic Test.

$$\int \frac{\sqrt[7]{3-\ln z^6}}{z}dz$$

Logarithmic Properties

$$\log_b b = 1$$

$$\log_b a^c = c\log_b a$$

$$a^{\log_a b} = b$$

$$\log a - \log b = \log\frac{a}{b}$$

$$\log a + \log b = \log ab$$

$$\int \frac{\sqrt[7]{3-\ln z^6}}{z}dz$$

$$\int \frac{\sqrt[7]{3-6\ln z}}{z}dz$$

$$\boxed{\begin{aligned} u &= 3-6\ln z \\ du &= -6\cdot\frac{1}{z}dz \end{aligned}}$$

$$\int \sqrt[7]{3-6\ln z}.\frac{-6dz}{-6z}$$

$$-\frac{1}{6}\int u^{1/7}du = -\frac{1}{6}\cdot\frac{u^{\frac{1}{7}+1}}{\frac{1}{7}+1} + C$$

$$-\frac{7}{48}(3-6\ln z)^{\frac{8}{7}} + C$$

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Differentiate implicitly the equation $x = \ln(x+y+1)$ to find the slope of the tangent line at any given point (x,y)

$$\frac{d}{dx}(x) = \frac{d}{dx}\left(\ln(x+y+1)\right)$$

$$1 = \frac{1}{x+y+1} \cdot \frac{d}{dx}(x+y+1)$$

$$1 = \frac{1}{x+y+1} \cdot 1 + y' + 0$$

$$1 + y' = x + y + 1$$

$$y' = x + y$$

Random Slope of $(-1, e^{-1})$

$$y' = -1 + \frac{1}{e}$$

Basic Integrals and Derivatives

$\frac{d}{dx}(\sin u) = \cos u du$	$\int \cos u du = \sin u + C$
$\frac{d}{dx}(\cos u) = -\sin u du$	$\int \sin u du = -\cos u + C$
$\frac{d}{dx}(\tan u) = \sec^2 u du$	$\int \sec^2 u du = \tan u + C$
$\frac{d}{dx}(\sec u) = \sec u \tan u du$	$\int \sec u \tan u du = \sec u + C$
$\frac{d}{dc}(\cot u) = -\csc^2 u du$	$\int \csc^2 du = -\cot u + C$
$\frac{d}{dc}(\csc u) = -\csc u \cot u du$	$\int \csc u \cot u du = -\csc u + C$

5.5 Derivative of exponential functions.

$$\frac{d}{dx}\left(2^x\right)$$

$$\frac{d}{dx}\left(e^u\right)=e^u du$$

$$a^{\log_a b}=b$$

Remember that if $a^u=x$

then $\log_a x=u$

Logarithm is another word

for exponent!!

$$\therefore \frac{d}{dx}\left(2^x\right)=\frac{d}{dx}\left[e^{\ln 2^x}\right]$$

Remember that $\ln(x)=\log_e(x)$

$$\therefore \frac{d}{dx}\left(2^x\right)=\frac{d}{dx}\left[e^{\ln 2^x}\right]=e^{\ln 2^x}\frac{d}{dx}\left(\ln 2^x\right)$$

$$=e^{\ln 2^x}\frac{d}{dx}\left[x\cdot\ln 2\right]$$

$$=3^{\log_e 2^x}\ln 2$$

$$=2^x\ln 2$$

$$\therefore \boxed{\frac{d}{dx}\left(a^u\right)=a^u\left(\ln a\right)}$$

5.5#4

Find the derivative of $y=5^{-2x}$

$$\frac{d}{dx}\left[a^u\right]=a^u\left(\ln a\right)du$$

$$\frac{d}{dx}\left[5^{-2x}\right]=5^{-2x}\left(\ln 5\right)(-2)$$

5.5#5

Find the derivative of the function $f(x)=x5^x$

$$f(x) = x5^x$$

$$f'(x) = \frac{d}{dx}(x) \cdot 5^x + x \cdot \frac{d}{dx}(5^x)$$

$$= 1 \cdot 5^x + x \cdot 5^x \cdot \ln 5 \cdot \frac{d}{dx}(x)$$

$$= 5^x + x \cdot 5^x \ln 5$$

$$= 5^x (1 + x \ln 5)$$

Integration of $\int a^u du$

$$\int a^\omega d\omega \qquad \boxed{\text{Remmember } a^u = e^{\ln a^u}}$$

$$\int e^\omega d\omega \qquad \int e^{\ln a^u} du$$

$$e^\omega + C \qquad \omega = \ln a^u$$

$$\omega = u \ln a$$

$$\frac{d\omega}{du} = \ln a$$

$$d\omega = \ln a du$$

$$\frac{d\omega}{\ln a} = du$$

$$\therefore \int e^\omega d\omega \qquad \int e^{\ln a^u} du \Rightarrow \int e^\omega \cdot \frac{d\omega}{\ln a}$$

$$= \frac{1}{\ln a} \int e^\omega d\omega$$

$$= \frac{1}{\ln a} e^\omega + C$$

$$\therefore \boxed{\int a^u du = \frac{1}{\ln a} a^u + C}$$

5.5#9

Find the integral of $\int 7^{-x} dx$

$$\int 7^{-x}\,dx$$

$$\boxed{\begin{array}{ll} a=7 & u=-x \\ \dfrac{du}{dx}=-1 & du=-dx \end{array}}$$

$$-\int 7^{u}\,du=-\int e^{\ln 7^{u}}\,du$$

$$=-\int e^{w}dw$$

$$\boxed{w=\ln 7^{u} \quad w=u(\ln 7) \quad \frac{dw}{du}=\ln 7 \quad dw=\ln 7\,du}$$

$$=-\frac{1}{\ln 7}\int e^{\ln 7^{u}}\,du$$

$$-\frac{1}{\ln 7}\cdot e^{\ln 7^{u}}+C$$

$$-\frac{1}{\ln 7}\cdot 7^{u}+C$$

#23 Exam Review

$$\int_{1}^{2} 6x^{2}4x^{3}dx$$

$$\boxed{\begin{array}{ll} u=x^{3} & du=3x^{2}2x \\ 2du=6x^{2}2x & \\ x=1,u=1 & x=2,u=8 \end{array}}$$

$$\int_{1}^{0} 4^{u}\,du=2\cdot\frac{1}{\ln 4}\cdot 4^{u}\Big]_{1}^{8}$$

$$=2\left[\frac{1}{\ln 4}\right]\left(4^{8}-4\right)$$

$$=\frac{2}{\ln 4}(65536-4)$$

$$=\frac{2}{\ln 4}(65532)$$

$$=\frac{2}{\ln 2^{2}}(65532)$$

$$=\frac{1}{\ln 2}(65532)$$

Exam Review #24

$$\int_1^2 \frac{6\ln x}{x}dx$$

$$u=\ln x \quad du=\frac{1}{x}dx$$

$$\int_0^{\ln 2} 6^u\,du$$

$$\frac{1}{\ln 6}6^u\Big]_0^{\ln 2}$$

$$\frac{1}{\ln 6}\left[6^{\ln 2}-6^0\right]$$

$$\frac{1}{\ln 6}\left[6^{\ln 2}-1\right]$$

$$\frac{6^{\ln 2}-1}{\ln 6}$$

5.7 Inverse Trigonometric Functions: Differentiation.

Key Concepts.

- For a function to have a derivative, it needs to be one-to-one.
- Trigonometric functions need to have their domain limited for them to be one-to-one.

If given a function such as y= arcsin u and asked to find the derivative, we proceed as follows.

$$y=\arcsin u$$

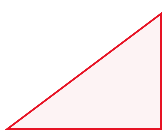
$$\sin y=u$$

$$\frac{d}{dx}(\sin y)=\frac{d}{dx}(u)$$

$$\cos y y'=du$$

$$y'=\frac{du}{\cos y}$$

If given $y=\arcsin e^{x^2}$ what is y'=?



$r=e^{x^2}$

$q=$

$$\sin y=e^{x^2}=\frac{e^{x^2}}{1}$$

$$\frac{d}{dx}(\sin y)=\frac{d}{dx}\left[e^{x^2}\right]$$

$$\cos y\cdot y'=e^{x^2}\cdot 2x$$

$$\cos y\cdot y'=2xe^{x^2}$$

$$y'=\frac{2xe^{x^2}}{\cos y}$$

$$y'=\frac{2xe^{x^2}}{\sqrt{1-e^{2x^2}}}$$

Since

$$\frac{d}{dx}(\arcsin u)=\frac{u'}{\sqrt{1-u^2}}$$

Then we can say that

$$\int\frac{du}{\sqrt{1-u^2}}=\arcsin u+C$$

5.8#4

Find the integral

$$\int\frac{6x+7}{\sqrt{1-x^2}}dx$$

We split it into two in order to fit the arcsin

$$\int\frac{6xdx}{\sqrt{1-x^2}}+\int\frac{7}{\sqrt{1-x^2}}dx$$

For $\int\frac{6xdx}{\sqrt{1-x^2}}$

$u=1-x^2 \quad du=-2x$

$$-3\int\frac{du}{\sqrt{u}}=-3\int u^{-\frac{1}{2}}du$$

For $\int\frac{7}{\sqrt{1-x^2}}dx$

$$=7\int\frac{1}{\sqrt{1-x^2}}dx$$

Combining the two yields

$C-6\sqrt{1-x^2}+7\sin^{-1}x$

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Exam Review #23

$$\int_1^2 6x^2 4^{x^3}\,dx$$

$u=x^3 \quad du=3x^2dx$

$x=1,u=1$

$x=2,u=8 \quad dx=\frac{du}{3x^2}$

$$2\int_1^8 4^u\,du$$

$$2\frac{1}{\ln 4}4^u\Big]_1^8$$

$$\frac{2}{\ln 2^2}\left[4^u\right]_1^8$$

$$\frac{1}{\ln 2}\left[4^8-4\right]$$

Exam I Review#24

$\int_1^2 \frac{6^{\ln x}}{x} dx$	$u = \ln x$	$du = \frac{1}{x} dx = \frac{dx}{x}$

$$\int_1^2 6^{\ln x} \frac{dx}{x}$$

$$\int 6^u du = \frac{1}{\ln 6} \cdot 6^u = \frac{1}{\ln y}\left[6^{\ln x}\right]_1^2$$

$$= \frac{1}{\ln y}\left[6^{\ln 2} - 6^{\ln 1}\right]$$

5.8 Inverse Trigonometric Functions: Integration

It is important to note that.

$\frac{d}{dx}[\arcsin u] = -\frac{d}{dx}[\arccos u]$	
$\frac{d}{dx}[\arctan u] = -\frac{d}{dx}[\operatorname{arccot} u]$	
$\frac{d}{dx}[\operatorname{arcsec} u] = -\frac{d}{dx}[\operatorname{arccsc} u]$	

- In order to master a formular, learning how to derive it a sure way.
- The composition of a function and it's inverses simply gives a variable as shown below

$$(f \circ g)(x) = x$$

For instance

$y = x^2 - 4$	$[-\infty, 0]$
$x = y^2 - 4$	$[-4, \infty]$
$\pm\sqrt{x+4} = y$	

$$g(x) = f^{-1}(x) = \left(-\sqrt{x+4}\right)$$

$$f(g(x)) = (g(x))^2 - 4$$
$$= \left(-\sqrt{x+4}\right)^2 - 4$$
$$= x + 4 - 4 = x$$

$\frac{d}{dx}[\arcsin u]$

$\sin y = \sin(\arcsin u)$	$y' = \frac{du}{\cos y}$
$\sin y = u$	$\sin y = u$
$\frac{d}{dx}[\sin y] = \frac{d}{dx} u$	$\sin y = \frac{u}{1}$
$\cos y \cdot y' = du$	

Find derivative of $y = \arcsin \ln(x^2)$

$$y' =$$
$$\sin y = \sin\left[\arcsin \ln\left(x^2\right)\right]$$
$$\sin y = \ln x^2$$
$$\frac{d}{dx}\left[\sin y\right] = \frac{d}{dx}\left[\ln x^2\right]$$
$$\cos y \cdot y' = \frac{2x}{x^2} = \frac{2}{x}$$

$$y' = \frac{2}{x \cos y}$$
$$but \cos y = \frac{\sqrt{1-\left(\ln x^2\right)^2}}{1}$$
$$y' = \frac{2}{x\sqrt{1-\left(\ln x^2\right)^2}}$$

Find the Derivative of arcsine (u)

$$\operatorname{arccsc} u$$
$$\csc y = \csc\left(\operatorname{arccsc} u\right)$$
$$\frac{d}{dx}\left[\csc y\right] = \frac{d}{dx}\left(u\right)$$
$$-\cot y \csc y \cdot y' = u'$$
$$y' = -\frac{u'}{\cot y \csc y}$$

Triange with;

$$opp = 1, hyp = u, adj = \sqrt{u^2 - 1}$$

$$= \frac{-u'}{\frac{\sqrt{u^2-1}}{1}\frac{u}{1}}$$
$$= -\frac{u'}{u\sqrt{u^2-1}}$$

Thursday, February 1, 2024

5.8 Inverse Trig Functions: Integration.

$\frac{d}{dx}\left(\arctan\frac{u}{a}\right)$	$=\frac{du}{a\left(\frac{\sqrt{a^2+u^2}}{a}\right)^2}$
$\tan y=\tan\left(\arctan\frac{u}{a}\right)$	$=\frac{adu}{a^2+u^2}$
$\frac{d}{dx}\left[\tan y\right]=\frac{d}{dx}\left[\frac{u}{a}\right]$	$\frac{d}{dx}\left(\arctan\frac{u}{a}\right)=\frac{adu}{a^2+u^2}$
Triangle: Opp=u,adj=a, hyp=$\sqrt{a^2+u^2}$	$\Rightarrow\int\frac{adu}{a^2+u^2}=\arctan\frac{u}{a}+C$
$\sec^2 y\cdot y'=\frac{1}{a}\cdot du$	$a\int\frac{du}{a^2+u^2}=\arctan\frac{u}{a}+C$
$y'=\frac{\frac{1}{a}du}{\sec^2 y}=\frac{du}{a\sec^2 y}$	$\int\frac{du}{a^2+u^2}=\frac{1}{a}\cdot\arctan\frac{u}{a}+C$

5.8#1

Find the indefinite integral.

$$\int \frac{16}{1+64x^2}\,dx$$

$$16\int \frac{dx}{1+64x^2} \left\} \boxed{\begin{aligned} &a=1, u=8x \\ &du=8dx \\ &dx=\frac{8}{du} \end{aligned}} \right\} \boxed{\frac{1}{a}\arctan\frac{u}{a}+C}$$

$$=\frac{16}{8}\cdot\frac{1}{1}\arctan\frac{8x}{1}+C$$

$$=2\arctan 8x + C$$

Proof

$$\frac{d}{dx}\left[16\arctan 8x + C\right]$$

$y = \arctan 8x + C$

$\tan y =$

Triangle:Opp=8x,adj=1

hyp=$\sqrt{64x^2+1}$

$y = \arctan 8x + C$

$\tan y = 8x$

$$\frac{d}{dx}\left[\tan y\right] = \frac{d}{dx}\left[8x\right]$$

$\sec^2 y \cdot y' = 8$

$$y' = (2)\frac{8}{\sec^2 y} = (2)\frac{8}{\left(\frac{\sqrt{64x^2+1}}{1}\right)^2}$$

$$=\frac{16}{64x^2+1}$$

5.8#3

Find the indefinite integral.

$$\int \frac{e^{2x}}{16+e^{4x}}\,dx$$

$a=4$	$u=e^{2x}$
$\frac{du}{dx}=2e^{2x}$	$du=2e^{2x}dx$

$$\frac{1}{2}\int \frac{du}{a^2+u^2} = \frac{1}{2}\cdot\frac{1}{a}\cdot\arctan\frac{u}{a}+C$$

$$=\frac{1}{2}\cdot\frac{1}{4}\arctan\frac{1}{4}e^{2x}+C$$

Derivation of the derivative of cosecant u.

$$y = \operatorname{arccsc}\frac{u}{a}$$

$$\csc y = \frac{u}{a}$$

Triangle opp=a,hyp=u

adj=$\sqrt{u^2 - a^2}$

$$\frac{d}{dx}(\csc y) = \frac{d}{dx}\left(\frac{u}{a}\right)$$

$$-\csc y \cot y \cdot y' = \frac{1}{a}\cdot du$$

$$y' = \frac{du}{a(-\csc y \cot y)}$$

$$= \frac{du}{-a\cdot\frac{u}{a}\cdot\frac{\sqrt{u^2-a^2}}{a}}$$

$$= -\frac{du}{\frac{\sqrt{u^2-a^2}}{a}}$$

$$\frac{d}{dx}\left(\operatorname{arccsc}\frac{u}{a}\right) = -\frac{adu}{u\sqrt{u^2-a^2}}$$

$$\int\frac{-adu}{u\sqrt{u^2-a^2}} = \operatorname{arccsc}\frac{u}{a}$$

$$-\frac{1}{a}\int\frac{-adu}{u\sqrt{u^2-a^2}} = -\frac{1}{a}\operatorname{arccsc}\frac{u}{a}$$

$$\int\frac{du}{u\sqrt{u^2-a^2}} = -\frac{1}{a}\operatorname{arcsec}\frac{|u|}{a} + C$$

5.8#5

$$\int_1^3 \frac{1}{x\sqrt{16x^2-4}}dx$$

$a=2$	
$u=4x$	$du=4dx$

$$\frac{4}{4}\int_1^3 \frac{4}{x4\sqrt{16x^2-4}}dx = \frac{1}{2}\operatorname{arcsec}\frac{|4x|}{2}\Bigg]_1^3$$

$$= \frac{1}{2}\left[\operatorname{arcsec}\frac{4(3)}{2} - \operatorname{arcsec}\frac{4(1)}{2}\right]$$

$$= \frac{1}{2}\left(\operatorname{arcsec}6 - \sec^{-1}2\right)$$

$$= \frac{1}{2}\left(\operatorname{arcsec}(6) - \frac{\pi}{2}\right)$$

Hyperbolic Functions.

Definitions of the Hyperbolic Functions.

$$\sinh x = \frac{e^x - e^{-x}}{2} \qquad \operatorname{csch} x = \frac{1}{\sinh x}, \qquad x \neq 0$$

$$\cosh x = \frac{e^x + e^{-x}}{2} \qquad \operatorname{sech} x = \frac{1}{\cosh x}$$

$$\tanh x = \frac{\sinh x}{\cosh x} \qquad \coth x = \frac{1}{\tanh x}, x \neq 0$$

Monday, February 5, 2024

Derivative of sinh x

$$y=\sinh x$$
$$y'=?$$
$$y=\frac{e^x-e^{-x}}{2}$$
$$y=\frac{1}{2}\left(e^x-e^{-x}\right)$$
$$y'=\frac{1}{2}\left[e^x(1)-e^{-x}(-1)\right]$$
$$y'=\frac{1}{2}\left[e^x+e^{-x}\right]$$
$$\therefore \int \cosh x=\sinh x+C$$

$\underline{\underline{\text{Proof of } \cosh^2-\cos^2 x=1}}$

$$\sinh^2 x-\cosh^2 x=-1$$
$$\sinh x=\frac{e^x-e^{-x}}{2}$$
$$\cosh x=\frac{e^x+e^{-x}}{2}$$
$$\frac{1}{4}\left(e^{2x}-2e^x e^{-x}+e^{-2x}\right)-\frac{1}{4}\left(e^{2x}+2e^x e^{-x}+e^{-2x}\right)\stackrel{?}{=}1$$
$$e^{2x}-2e^x e^{-x}+e^{-2x}-\left(e^{2x}+2e^x e^{-x}+e^{-2x}\right)=4$$
$$e^{2x}-2+e^{-2x}-e^{2x}-2-e^{-2x}=4$$
$$-4=4$$

Which was an error but also able to prove that

$$\cosh^2 x-\sinh^2 x=1$$

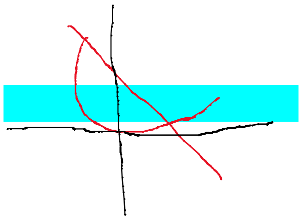
5.9#4

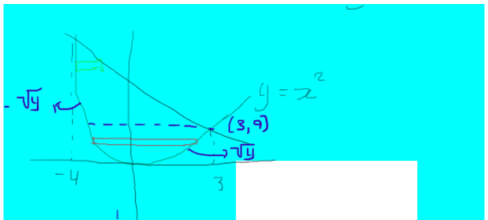
Find the derivative of $\sinh 6x$

$$y'=\cosh 6x\cdot 6$$
$$y'=6\cosh 6x$$

7.1#3

Consider the following. $Y=x^2$, $y=12-x$





$$\int_0^9 (\sqrt{y} - (-\sqrt{y})) dy + \int_9^{16} [(12 - y) - (-\sqrt{y})] dy$$

Find limits $x^2 = 12 - x$ $(x+4)(x-3) = 0$ $x = -4, 3$ $\int_{-4}^{3} 12 - x - (x^2) dx$ $12x - \frac{1}{2}x^2 - \frac{1}{3}x^3 \Big]_{-4}^{3}$ $12(3) - \frac{1}{2}(3)^2 - \frac{1}{3}(3)^3 - \left[12(-4) - \frac{1}{2}(-4)^2 - \frac{1}{3}(-4)^3\right]$ $36 - \frac{9}{2} - 9 - \left(-48 - 8 + \frac{64}{3}\right)$	$\int_0^9 2\sqrt{y} dy + \int_9^{16} \left(12 - y + \sqrt{y}\right) dy$ $2\int_0^9 y^{1/2} dy$ $2 \cdot \frac{2}{3} y^{3/2} \Big]_0^9 + 12 - \frac{1}{2}y^2 + \frac{2}{3}y^{3/2} \Big]_9^{16}$ $12(16) - \frac{1}{2}(16)^2 + \frac{2}{3}(4^2)^{3/2} - \left[12(9) - \frac{1}{2}(81) + \frac{2}{3}(3^2)^{3/2}\right]$ $12(16) - \frac{1}{2} \cdot 16 \cdot 16 + \frac{2}{3} \cdot 64 - 12(9) + \frac{81}{2} - 18$ $12(16-9) + \frac{81}{2} - 146$	

Tuesday, February 6, 2024

5.5#6

$h(x) = \log_3 \frac{x\sqrt{x-6}}{3}$ $= \log_3 x\sqrt{x-6}$ $= \log_3 x\sqrt{x-6} - 1$ $= \log_3 x + \log_3 \sqrt{x-6} - 1$ $= \log_3 x + \log_3 (x-6)^{\frac{1}{2}} - 1$ $h(x) = \log_3 x + \frac{1}{2}\log_3 (x-6) - 1$ $= \frac{\ln x}{\ln 3} + \frac{1}{2}\left[\frac{\ln(x-6)}{\ln 3}\right] - 1$	$h'(x) = \frac{1}{\ln 3} \cdot \frac{1}{x} + \frac{1}{\ln 9} \cdot \frac{1}{x-6}$ $h'(x) = \frac{1}{\ln 3^3} + \frac{1}{\ln 9^{(x-6)}}$	

5.5#7

$\ln y = \ln x^{\frac{3}{x}}$ $\ln y = \frac{3}{x}\ln x$ $\frac{1}{y}\cdot y' = 3\cdot\frac{d}{dx}\left[\frac{1}{x}\cdot\ln x\right]$ $\frac{y'}{y} = 3\left[-\frac{1}{x^2}\cdot\ln x + \frac{1}{x}\cdot\frac{1}{x}\right]$	$\frac{y'}{Y} = 3\left[-\frac{1}{x^2}\ln x + \frac{1}{x^2}\right]$ $y' = y\left\{\frac{3}{x^2}[1-\ln x]\right\}$ $y' = x^{3/x}\left\{\frac{3}{x^2}[1-\ln x]\right\}$	

5.5#12

$\int_{-2}^{2} 36^{\frac{1}{2}x}dx$ $\int a^u du = \frac{1}{\ln a}a^u + C$ $u=\frac{1}{2}x$ $\quad\frac{du}{dx}=\frac{1}{2}$ $du=\frac{1}{2}dx$ if x=-2, u=-1 x=2, u=1 $\int_{-1}^{1}36^u 2du$ $2\int_{-1}^{1}36^u du$ $2\left[\frac{1}{\ln 36}36^u\right]_{-1}^{1}$ $\frac{2}{\ln 36}\left[36^1-36^{-1}\right]$	$\frac{1}{\ln 6}\left(36-\frac{1}{36}\right)$ $\frac{1}{\ln 6}\left(\frac{36^2-1}{36}\right)$ $\frac{36^2-1}{36\ln 6}$ $\boxed{\frac{36^2-1}{\ln 6^{36}}}$	

5.7#9

$\int_1^2 \frac{2x-3}{\sqrt{4x-x^2}}$ $\int\frac{2x-3}{\sqrt{4x-x^2}}dx$ $\int\frac{2x}{\sqrt{4x-x^2}}dx-3\int\frac{dx}{\sqrt{4x-x^2}}$ $-\int\frac{(-1)2x}{\sqrt{4x-x^2}}dx-3\int\frac{dx}{\sqrt{4x-x^2}}$ $-\int\frac{-2x+4-4}{\sqrt{4x-x^2}}dx-3\int\frac{dx}{\sqrt{4x-x^2}}$ $-\int\frac{4-2x}{\sqrt{4x-x^2}}dx+4\int\frac{dx}{\sqrt{4x-x^2}}-3\int\frac{dx}{\sqrt{4x-x^2}}$	$\int\frac{dx}{\sqrt{4x-x^2}}-\int\frac{4-2x}{\sqrt{4-x^2}}dx$ $\int\frac{dx}{\sqrt{2^2-(x-2)^2}}-\int\frac{4-2x}{\sqrt{4-x^2}}dx$ $=\arcsin\frac{(x-2)}{a}\Big]_1^2-\int\frac{du}{u}$ $=\arcsin\frac{(x-2)}{a}\Big]_1^2-2\sqrt{u}\Big]_1^2$ $=\arcsin\frac{(x-2)}{a}\Big]_1^2-2\sqrt{u}\Big]_1^2$	$=\sin^{-1}(0)-\arcsin\left(-\frac{1}{2}\right)-2\left[2-\sqrt{3}\right]$ $=0-\left(-\frac{\pi}{6}\right)-4+2\sqrt{3}$ $=\frac{\pi}{6}-4+2\sqrt{3}$

5.9 Hyperbolic Functions.

Proof of $\cosh^2 x - \sinh^2 x = 1$

$$\left(\frac{e^x + e^{-x}}{2}\right)^2 - \left(\frac{e^x - e^{-x}}{2}\right)^2 = 1$$

$$\frac{e^{2x} + 2 \cdot e^x e^{-x} + e^{-2x}}{4} - \frac{e^{2x} - 2e^x e^{-x} + e^{-2x}}{4} = 1$$

$$\frac{e^{2x} + 2 + e^{-2x} - e^{2x} + 2 - e^{-2x}}{4} = 1$$

$$\frac{4}{4} = 1$$

$$1 = 1$$

Proof of $\frac{d}{dx}(\sinh u) = \cosh u$

$$\frac{d}{dx}(\sinh u) = \frac{d}{dx}\left[\frac{e^u - e^{-u}}{2}\right]$$

$$\frac{1}{2} \cdot \frac{d}{dx}\left[e^u - e^{-u}\right]$$

$$= \frac{1}{2}\left[e^u du - e^{-u}(-1) du\right]$$

$$= \cosh u$$

Proof of sinhx=$\ln\left(x + \sqrt{x^2 + 1}\right)$

$$y = \sinh x = \frac{e^x - e^{-x}}{2}$$

$$x = \sinh y = \frac{e^y - e^{-y}}{2}$$

$$2x = e^y - \frac{1}{e^y}$$

$$2x = \frac{e^{2y} - 1}{e^y}$$

$$2xe^y = e^{2y} - 1$$

$$0 = e^{2y} - 2xe^y - 1$$

$$e^y = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(1) \pm \sqrt{(-2x)^2 - 4(1)(-1)}}{2(1)}$$

$$= e^y = \frac{2x \pm \sqrt{4x^2 + 4}}{2}$$

$$= \frac{2x \pm 2\sqrt{x^2 + 1}}{2}$$

$$\frac{2\left(x \pm \sqrt{x^2 + 1}\right)}{2} = e^y$$

$$e^y = x + \sqrt{x^2 + 1}$$

$$\ln e^y = \ln\left(x + \sqrt{x^2 + 1}\right)$$

$$y = \ln\left(x + \sqrt{x^2 + 1}\right)$$

derivative of $\left[\sinh^{-1} u\right] = \dfrac{u'}{\sqrt{u^2+1}}$ $\dfrac{d}{dx}\left[\sinh^{-1} u\right] = \dfrac{d}{dx}\left[\ln\left(u+\sqrt{u^2+1}\right)\right]$ $= \dfrac{1du + \frac{1}{}}{}$ $\boxed{\text{NOT FINISHED}}$		

Wednesday, February 7, 2024

7.2#4

Find the volume of the solids generated by revolving the region bounded by the graphs of the equations about the given lines.

$y = \sqrt{x}$ $y = 0$ $x = 3$ $V = \pi\int_0^3 \left(\sqrt{x}-0\right)^2 dx$ $= \pi\int_0^3 xdx$ $= \pi\left[\dfrac{x^2}{2}\right]_0^3 = \dfrac{9\pi}{2}$		

$\pi\int_0^{\sqrt{3}} (9-y)^2 - (9-3)^2\, dx$ $\pi\int_0^{\sqrt{3}} 81 - 18y + y^2 - 9dx$ $= \dfrac{144\pi\sqrt{3}}{5}$

7.3 Shell Method.

$\int_0^2$ The volume of the shell is given by $r-\frac{w}{2}=$ radius of hole $r+\frac{w}{2}=$ radius of outside cylinder $V=\pi r^2 h$ $V_0=\pi\left(r+\frac{w}{2}\right)^2 h$ $V_H=\pi\left(r-\frac{w}{2}\right)^2 h$	Volume of Shell V_0-V_H $\pi\left(r+\frac{w}{2}\right)^2 h-\pi\left(r-\frac{w}{2}\right)^2 h$ $\pi h\left[r^2+rw+\frac{w^2}{4}-\left(r^2-rw+\frac{w^2}{4}\right)\right]$ $\pi h\left[r^2+rw+\frac{w^2}{4}-r^2+rw-\frac{w^2}{4}\right]$ $=\pi h 2rw$ $\therefore V_{\text{SHELL}}=2\pi\int_0^2 \overbrace{r(x)}^{\text{Radius}}\overbrace{h(x)}^{\text{Height}}\overbrace{dy}^{\text{Thickness}}$	For our example we have $2\pi\int_0^2 y\left(4-y^2\right)dy$ $2\pi\int_0^2\left(4y-y^3\right)dy$ $2\pi\left[4\cdot\frac{1}{2}y^2-\frac{1}{4}y^4\right]_0^2$ $2\pi\left[2\cdot 2^2-\frac{1}{4}\cdot 2^4-\left(2\cdot 0^2-\frac{1}{4}\cdot 0\right)\right]$ $2\pi[8-4]$ $=\boxed{8\pi}$

Monday, February 12, 2024

7.1#7

Consider the following functions. $\boxed{\begin{array}{l} f(x)=5\sin x \\ g(x)=5\cos 2x \\ -\frac{\pi}{2}\le x\le\frac{\pi}{6}\end{array}}$ Sketch the region bounded by the graphs $\int_{-\frac{\pi}{2}}^{\frac{\pi}{6}}(5\cos 2x-5\sin x)dx$ $5\int_{-\frac{\pi}{2}}^{\frac{\pi}{6}}(\cos 2x-\sin x)dx=5\int_{-\pi}^{\frac{\pi}{3}}\cos u\frac{du}{2}$ $\boxed{\begin{array}{l} u=2x \\ du=2dx \\ \frac{du}{2}=dx\end{array}}$ $\boxed{\begin{array}{l} \text{if x=-}\frac{\pi}{2},\ u=2\left(-\frac{\pi}{2}\right)=-\pi \\ \textit{if}\ \text{x=}\frac{\pi}{6},\ u=\left(\frac{\pi}{6}\right)=\frac{\pi}{3}\end{array}}$	$=\frac{5}{2}\int_{-\pi}^{\frac{\pi}{2}}\cos u du$ $\frac{5}{2}\cdot\sin u\Big]_{-\pi}^{\frac{\pi}{3}}$ $\frac{5}{2}\left(\sin\frac{\pi}{3}-\sin(-\pi)\right)$ $=\frac{5}{2}\left[\frac{\sqrt{3}}{2}-0\right]=\frac{5\sqrt{3}}{4}$	

Tuesday, February 13, 2024

7.1#7

$$\int_0^2 \left[(-x+4)-\left(x^2-9\right)\right]dx$$
$$\int_{-\pi/2}^{\pi/6} (9\cos 2x - 9\sin x)\,dx$$
$$9\int_{-\pi/2}^{\pi/6} (\cos 2x - \sin x)\,dx$$
$$\ldots$$

7.1#9

$$F(x) = \int_0^x \left(\frac{1}{2}t + 9\right)dt$$
$$= \frac{1}{4}t^2 + 9t\Big]_0^x$$
$$= \frac{1}{4}x^2 + 0x - (0+0)$$
$$= \frac{1}{4}x^2 + 9x$$
$$F(6) = \frac{1}{4}(6)^2 + 9\cdot 6$$
$$= 9 + 54$$
$$= 63$$

7.1#11

$$2m\left[2\left[\int_0^5 \left(1-\frac{1}{3}\sqrt{5-x}\right)dx + \int_5^{11/2} (1-0)\,dx\right]m^2\right]$$

Exam Review ma2414x2reveopen.pdf.#2

<u>With Disk Method</u> Find the volume of the solid generated by revolving the region bounded by the given lines and curves about the x-axis. $\boxed{y=\sqrt{\sin 6x}}$ $\boxed{y=0}$ $\boxed{0\le x\le \frac{\pi}{6}}$ After drawing the sketch, with atleast two representative rectangles $V=\pi\int r^2h$ $=\pi\int_0^{\pi/6}\left(\sqrt{\sin 6x}-0\right)^2 dx$ $=\frac{\pi}{6}\int_0^{\pi/2}(\sin 6x)(6)dx=\frac{\pi}{6}\int_0^{\pi}\sin u du$ $=\frac{\pi}{3}$	<u>With Shell Method</u> $V=2\pi\int \overbrace{p}^{\substack{\text{Height}\\ \text{of}\\ \text{Shell}}}\ \overbrace{r}^{\substack{\text{Radius}\\ \text{of}\\ \text{Shell}}}\ \overbrace{w}^{\substack{\text{Thickness}\\ \text{of}\\ \text{Shell}}}$ $V=2\pi\int_0^1 y\left[\frac{1}{6}\arcsin\left(y^2\right)-\frac{1}{6}\arcsin\left(y^2\right)\right]dy$	

Derivation of the arc length Formular.

$\sum\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$ $\lim_{x\to\infty}\sum_{i=1}^{n}\sqrt{\Delta x_i^2+\Delta y_i^2}\cdot\frac{\Delta y}{\Delta x}$ $\sum_{i=i}^{n}\sqrt{\Delta x_i^2+\Delta y_i^2}\,\frac{\Delta x}{\sqrt{\Delta x^2}}$ $\sum_{i=i}^{n}\frac{\sqrt{\Delta x_i^2+\Delta y_i^2}}{\sqrt{\Delta x^2}}\Delta x$ $\sum\sqrt{\frac{\Delta x_i^2}{\Delta y_i^2}+\frac{\Delta y_i^2}{\Delta x_i^2}}$ … $\sum\sqrt{1-\left[f'(x)\right]^2}\Delta x$		

Derivation of the Surface Area of Revolution.

$$Area=2\pi\int rLdx=2\pi\int r\sqrt{1+\left[f'(x)\right]^2}dx$$

Since L is the arclength is w

Wednesday, February 14, 2024

Exam Review #4

$\int_1^x \sqrt{t^2-1}dt$ Trig sub let t=$\sec\theta$ $t^2=\sec^2\theta$ $t^2-1=\tan^2\theta$ $\sqrt{\tan^2\theta}=\tan\theta$ $\frac{dt}{d\theta}=\sec\theta\tan\theta$ $\int \tan\theta\sec\theta\tan\theta d\theta$ $\int \tan^2\theta\sec\theta d\theta$ $\int(\sec^2\theta-1)(\sec\theta)d\theta$ $\int(\sec^3\theta-\sec\theta)d\theta$	$\int \sec^3\theta d\theta-\int\sec\theta d\theta$ $\int\sec^3\theta d\theta=\sec\theta\tan\theta-\int\tan\theta\sec\theta\tan\theta d\theta$ $=\sec\theta\tan\theta-\int\tan^2\theta\sec\theta d\theta+C$ $\int\sec^2\theta\sec\theta d\theta$ $\int udv=uv-\int vdu$ $\boxed{u=\sec\theta \quad dv=\sec^2\theta d\theta \quad du=\sec\theta\tan\theta \quad v=\int\sec^2\theta d\theta=\tan\theta}$ $2\int\tan^2\theta\sec\theta d\theta=\sec\theta\tan\theta+C$ $\int\tan^2\theta\sec\theta d\theta=\frac{1}{2}\sec\theta\tan\theta-\frac{1}{2}\int\sec\theta d\theta$	$\int_1^x\sqrt{t^2-a}dt=\int\tan^2\theta\sec\theta d\theta$ $=-\frac{1}{2}\int\sec\theta d\theta+\frac{1}{2}\sec\theta\tan\theta$ $=-\frac{1}{2}\ln\lvert\sec\theta+\tan\theta\rvert+\frac{1}{2}\sec\theta\tan\theta$ $\frac{t}{1}=\sec\theta$ $\boxed{\text{Triangle:Opp=}\sqrt{t^2-1}, adj=1, hyp=t}$ $=-\frac{1}{2}\ln\left\lvert\frac{t}{1}+\frac{\sqrt{t^2-1}}{1}\right\rvert+\frac{1}{2}t\cdot\sqrt{t^2-1}$ $=\frac{1}{2}t\sqrt{t^2-1}\cdots+\sqrt{t^2-1}$ $\boxed{y=\frac{1}{2}x\sqrt{x^2-1}-\frac{1}{2}\ln\lvert x+\sqrt{x^2-1}\rvert}$

7.4#9

Set up the integral.

$y=\frac{1}{2}x^3$ SA=$2\pi\int rLdx$ $SA=2\pi\int rLdx$ $=2\pi\int_0^3\left(\frac{1}{3}x^3-0\right)$ $=2\pi\int_0^3\left(\frac{1}{3}x^3-0\right)\sqrt{1+(x^2)^2}dx$ $=\frac{2\pi}{3}\int_0^3 x^3\sqrt{1+x^4}dx$ $\boxed{u=1+x^4 \quad \frac{du}{dx}=4x^3}$	$\frac{2\pi}{3}\int x^3\sqrt{u}\frac{du}{4x^3}$ $\frac{\pi}{6}\int u^{1/2}du$ $\boxed{\text{x=0,u=1+}0^4=1 \quad x=3, u=1+3^4=82}$ $\frac{\pi}{6}\left[82^{3/2}-1^{3/2}\right]$ $\boxed{\frac{\pi}{6}\left[82^{3/2}-1\right]}$	

7.2#8

Find the volume revolved about the x-axis.

$\boxed{\begin{array}{l} y=\cos 2x \\ y=0 \\ x=0 \\ x=\dfrac{\pi}{4} \end{array}}$ $\boxed{\begin{array}{l}\text{After sketching the solid} \\ \text{of revolution with atleast} \\ \text{two representative rectangles.}\end{array}}$ $V=\pi\int r^2dx$ $=\pi\int_0^{\pi/4}(\cos 2x-0)^2$ $=\pi\int_0^{\pi/4}(\cos 2x-0)^2\,dx$ $=\pi\int_0^{\pi/4}\cos^2 2xdx$ $\boxed{\begin{array}{l}\cos(A+B)=\cos A\cos B-\sin A\sin B \\ \cos(A+A)=\cos^2 A-\sin^2 A \\ \cos(2A)=\cos^2 A-(1-\cos^2 A) \\ \cos 2A=2\cos^2 A-1 \\ \cos 2A=2\cos^2 A-1 \\ \dfrac{1+\cos 2A}{2}=\cos^2 A\end{array}}$	$=\pi\int_0^{\pi/4}\left(\dfrac{1+\cos 4x}{2}\right)dx$ $=\pi\int_0^{\pi/4}\dfrac{1+\cos 4x}{2}dx$ $=\dfrac{\pi}{2}\int_0^{\pi/4}(1+\cos 4x)dx$ $\boxed{\begin{array}{lll} u=4x & x=0 & u=0 \\ du=4dx & & x=\dfrac{\pi}{4}, \\ \dfrac{du}{4}=dx & & \end{array}}$ $\dfrac{\pi}{2}\left[\dfrac{\pi}{4}-0+\dfrac{1}{4}\sin u\right]_0^{\pi}$ $=\dfrac{\pi^2}{8}$	

Thursday, February 15, 2024

7.2#7

$-x^2+2x+6=x^2+2$ $0=2x^2-2x-4$ $0=2\left(x^2-x-2\right)$ $x^2-x-2=0$ $(x-2)(x+1)=0$ $\pi\int_0^2\left[\left(-x^2+2x+6\right)^2-\left(x^2+2\right)^2\right]dx+\pi\int_2^3\left[\left(x^2+2-0\right)^2-\left(-x^2+2x+6\right)^2\right]$ $\pi\int_0^2\left(-4x^3-12x^2+24x+32\right)dx+\int_2^3\left(4x^3+12x^2-24x-32\right)dx$ $\pi\left[-2^4-4\cdot 2^3+12\cdot 2^2+32\cdot 2-0\right]+\pi\left[\begin{matrix}3^4+4\cdot 3^3-12\cdot 3^2-32\cdot 3- \\ \left(2^4+4\cdot 2^3-12\cdot 2^2-32\cdot 2\right)\end{matrix}\right]$ $\pi\left[-16-32+48+64\right]+\pi\left[\begin{matrix}81+108-108-96- \\ \left(16+32-48-64\right)\end{matrix}\right]$ $=113\pi$		

7.2#6

Washer Method	Disk Method	
$y=3-x$ $x=3-y$ $\pi\int_0^2\left[(3-0)^2-\left(3-(3-y)\right)^2\right]$ $\pi\int_0^2\left[3^2-\left[3-3+y\right]^2\right]dy$ $\pi\int_0^2\left(9-y^2\right)dy$ $\pi\left[9y-\frac{1}{3}y^3\right]_0^2=\pi\left[9\cdot 2-\frac{1}{3}\cdot 2^3\right]$ $\pi\left[18-\frac{8}{3}\right]=\frac{46\pi}{3}$	$2\pi\int_0^1 prw+2\pi\int_1^3 prw$ $2\pi\int_0^1(2-0)(3-x)dx+2\pi\int_1^3(3-x-0)(3-x)dx$ $2\pi\int_0^1 2(3-x)dx+2\pi\int_1^3(3-x)^2dx$ $2\pi\int_0^1(6-2x)dx+2\pi\int_1^3\left(9-6x+x^2\right)dx$ $2\pi\left[6x-x^2\right]_0^1+\left[9x-3x^2+\frac{1}{3}x^3\right]_1^3$ $2\pi[6-1-0]+\left(27-27+9-\left(9-3+\frac{1}{3}\right)\right)$ $2\pi\left[5+\left(5+2\frac{2}{3}\right)\right]$ $\frac{46\pi}{3}$	

7.4#5

Find the arclength of the graph over the indicated interval.

7.4#9

<table>
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$SA = 2\pi \int rL$

$= 2\pi \int_a^b r\sqrt{1+(y')^2}\,dx$

$= 2\pi \int_0^3 \left(\frac{1}{3}x^3 - 0\right)\sqrt{1+\left(x^2\right)^2}$

$= 2\pi \int_0^3 \left(\frac{1}{3}x^3\sqrt{1+x^4}\right)dx$

$= \frac{2\pi}{3}\int_0^3 \left(x^3\sqrt{1+x^4}\,dx\right)$

$\boxed{u = 1+x^4 \quad \frac{du}{dx} = 4x^3 \quad \frac{du}{4x^3} = dx}$

$\frac{2\pi}{3}\int \cancel{x^3}\sqrt{u}\,\frac{du}{4\cancel{x^3}}$

$\frac{\pi}{6}\int \sqrt{u}\,du = \frac{\pi}{6}\int u^{1/2}du$

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$\boxed{x = 0 \Rightarrow u = 1+0^4 = 1 \quad x = 3 \Rightarrow u = 1+81 = 82}$

$\frac{\pi}{6}\left[\frac{2}{3}u^{3/2}\right]_1^{82}$

$\frac{\pi}{9}\left[82^{3/2} - 1^{3/2}\right]$

$\frac{\pi}{9}\left(82^{3/2} - 1\right)$

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Thursday, February 22, 2024

Exam 1 Revision.

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$\int_1^{\sqrt{2}} x9^{x^2}dx$

$\boxed{u = x^2 \quad \frac{du}{dx} = 2x \quad du = 2xdx}$

$\int_1^2 9^u \frac{du}{2}$

$\boxed{\text{if x=1, u=}1^2 = 1 \quad \text{if x=}\sqrt{2}, u = \left(\sqrt{2}\right)^2}$

$\int a^u du = \frac{1}{\ln a}a^u + C$

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Exam 1 Review.

$y=-\cot^{-1}\left(\dfrac{8x}{3}\right)$ $y=\operatorname{arccot}\left(\dfrac{8x}{3}\right)$ $\cot y=\left(\dfrac{8x}{3}\right)$ $\boxed{\text{Triangle: Opp=3, Adj=8x, hyp=}\sqrt{64x^2+9}}$ $-\csc^2 y\cdot y'=\dfrac{8}{3}$ $y'=\dfrac{8/3}{-\csc^2 y}=\dfrac{-8}{3\cdot\dfrac{64x^2+9}{9}}$	$=\dfrac{-24}{64x^2+9}$ $\therefore \dfrac{24}{64x^2+9}$	

Exam 1 Review

$\int_0^{\frac{1}{3}}\dfrac{5xdx}{\sqrt{16-x^4}}$ $5\int_0^{1/3}\dfrac{xdx}{\sqrt{16-x^4}}$ $\boxed{a^2=16\quad u^2=x^4 \quad a=4 \quad \boxed{u=x^2}}$ $\int\dfrac{du}{a^2-u^2}=\sin^{-1}\dfrac{u}{a}+C$ $\int\dfrac{du}{u\sqrt{u^2-a^2}}=\dfrac{1}{a}\sec^{-1}\dfrac{\lvert u\rvert}{a}+C$ $\int\dfrac{du}{a^2+u^2}=\dfrac{1}{a}\tan^{-1}\dfrac{u}{a}+C$	$5\int\dfrac{du/2}{\sqrt{a^2-u^2}}$ $\dfrac{5}{2}\int\dfrac{du}{\sqrt{a^2-u^2}}=\dfrac{5}{2}\arcsin\dfrac{x^2}{4}+C$ $=\dfrac{5}{2}\left[\sin^{-1}\dfrac{x^2}{4}\right]_0^{\frac{1}{3}}$ $\dfrac{5}{2}\left[\sin^{-1}\dfrac{1/9}{4}-\sin^{-1}0\right]$ $=\dfrac{5}{2}\sin^{-1}\dfrac{1}{36}$	

Exam 1 Review

$\int \frac{dt}{t^2+8t+20}$ $\int \frac{dt}{t^2+8t+16+4}$ $\int \frac{dt}{(t+4)^2+2^2}$ $=\frac{1}{2}\tan^{-1}\left(\frac{t+4}{2}\right)+C$		

Exam 1 Review

$y=7^{\cos\theta}$ $y=a^u$ $\boxed{y'=\ln u\left(a^u\right)+C}$ $=(\ln 7)7^{\cos\theta}(-\sin\theta)$		

Exam 1 Question Review.

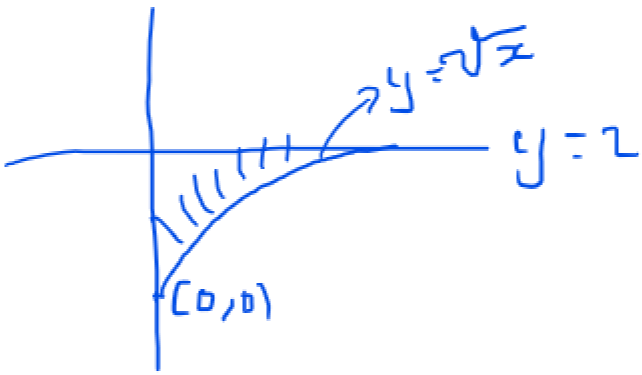
$\ln y=\ln x^{10\sin x}$ $\frac{d}{dx}[\ln y]=\frac{d}{dx}[10\sin x\ln x]$ $\frac{1}{y}\cdot y'=10\left[\cos x\cdot\ln x+\frac{1}{x}\sin x\right]$ $y'=10x^{10\sin x}\left(\cos x\ln x+\frac{\sin x}{x}\right)$		

Exam 1 Question Review

$\int\cosh^2\left(4-\frac{x}{4}\right)dx$ $\int\cosh^2 u du=-\coth u+C$ $\int\operatorname{csch}^2 u\cdot(-4du)$ $-4\int\operatorname{csch}^2 udu$ $-4\left(-\coth\left(4-\frac{1}{4}x\right)\right)+C$ $4\coth\left(4-\frac{1}{4}x\right)+C$	$2\pi\int_0^{\pi/4}\sin x\sqrt{1+\cos^2 x}dx$	

Monday, February 26, 2024

$V=\pi\int_0^4 r^2dx$ $=\pi\int_0^4\left[\left(2-\sqrt{x}\right)^2\right]dx$ By Shell Method. $V=2\pi\int rpw$ $=2\pi\int_0^2(2-y)\left(y^2-0\right)dy$	$V=\pi\int_0^4 r^2dx$ $=\pi\int_0^4\left[\left(2-\sqrt{x}\right)^2\right]dx$ By Shell Method. $V=2\pi\int rpw$ $=2\pi\int_0^2(2-y)\left(y^2-0\right)dy$



Random example in class

$\int\frac{1+x}{x^2+1}dx$ $\int\frac{1}{x^2+1}dx+\int\frac{x}{x^2+1}dx$ $=\frac{1}{a}\arctan\frac{u}{a}+C$ $u^2=x^2$ $u=x$ $a^2=1$ $a=1$ $\arctan x+\frac{1}{2}\ln\left(x^2+1\right)+C$	$\arctan x+\ln\sqrt{x^2+1}+C$ Triangle:hyp=$\sqrt{x^2+1}, adj=1, opp=x$	

Random example in class

$\int \frac{1}{\sqrt{2x-x^2}}$ $\int \frac{du}{u\sqrt{u^2-a^2}} = \frac{1}{a}\sec^{-1}\frac{\lvert u \rvert}{a} + C$ $\int \frac{du}{\sqrt{a^2-u^2}} = \arcsin\frac{u}{a} + C$ $2x-x^2 = -1\left(x^2-2x+1-1\right) = -1\left(x^2-2x+1\right)+1$ $=1-\left(x-1\right)^2$ $\boxed{\begin{array}{ll} a^2=1 & u^2=\left(x-1\right)^2 \\ a=1 & u=x-1 \end{array}}$ $\int \frac{du}{\sqrt{a^2-u^2}} = \arcsin\frac{u}{a} + C$		

Random example in class.

$\int \frac{2x}{x^2+2x+1}dx$ $u = x^2+2x+1$ $\frac{du}{dx} = 2x+2$ $\int \frac{2x+0}{x^2+2x+1}dx = \int \frac{2x+2-2}{x^2+2x+1}dx$ $\int \frac{2x+2}{x^2+2x+1}dx - 2\int \frac{1}{x^2+2x+1}dx$ $\ln\left\lvert x^2+2x+1 \right\rvert - 2\int \frac{du}{u^2}$ $\boxed{u=x+1}\ldots$		

Random example in class

$\int \frac{1}{1+\sin x} \cdot \frac{1-\sin x}{1-\sin x} dx$ $\int \frac{1-\sin x}{1-\sin^2 x} dx$ $\int \frac{1-\sin x}{\cos^2 x} dx$ $\int \left(\frac{1}{\cos^2 x} - \frac{\sin x}{\cos x \cos x} \right) dx$ $\int (\sec^2 x - \tan x \sec x) dx$ $= \tan x - \sec x + C$		

Random example in class

$\int x^2 \cdot \sin x dx$ $\int x \sin x^2 dx$ $\boxed{u = x^2 \quad du = 2x dx \quad \frac{du}{2} = dx}$ Integration by parts works when you have two unrelated functions. It's origin is in derivative of the product. As shown below $\frac{d}{dx}(u \cdot v) = \frac{du}{dx} \cdot v + \frac{dv}{dx} \cdot u$ $\frac{d}{dx}(u \cdot v) = u' \cdot v + v' \cdot u$ $= v \cdot du + u \cdot dv$ $\int \frac{d}{dx}(u \cdot v) = \int (v \cdot du + u \cdot dv) dx$ $uv = \int vdu + \int udv$ $uv - \int vdu = \int udv$	Use LIATE in determing the order of priority for the u. L logarithmic I Inverse trigonometric functions. A Algebraic functions. T Trigonometric functions E exponension functions. $\int x^2 \cdot \sin x dx$ $\boxed{u = x^2 \quad dv = \sin x dx \quad v = \int dv = \int \sin x dx \quad v = -\cos x}$ $= uv - \int vdu$ $= x^2 \cdot (-\cos x) - \int -\cos x (2x) dx$ $= -x^2 \cos x + 2\int x \cos x dx$ $\boxed{u = x, \quad du = 1 \quad dv = \cos x \quad v = \sin x}$ $= -x^2 \cos x + 2\left[x \sin x - \int \sin x dx \right]$ $= -x^2 \cos x + 2[x \sin x - \cos x]$ $= -x^2 \cos x + 2x \sin x - 2\cos x$	

Tuesday, February 27, 2024

8.3#5

$$\int xe^{8x}dx$$

$$\boxed{LI\boxed{A}TE}$$

$$\boxed{\begin{array}{ll} u = x & du = dx \\ dv = e^{8x}dx & v = \frac{1}{8}e^{8x} \end{array}}$$

$$\boxed{\int udv = uv - \int vdu}$$

$$x \cdot \frac{1}{8}e^{8x} - \int \frac{1}{8}e^{8x}dx$$

$$\boxed{u = 8x, \qquad du = 8dx}$$

$$-\frac{1}{8}\int e^{u}\frac{du}{8}$$

$$\frac{1}{8}xe^{8x} - \frac{1}{64}e^{8x} + C$$

8.2#10

$$\int \frac{xe^{2x}}{(2x+1)^2}\,dx$$

$$u = \frac{x}{(2x+1)^2} \qquad dv = e^{2x}dx$$

$$du = \frac{1\cdot(2x+1)^2 - \left(2(2x+1)'\cdot 2\cdot x\right)}{\left((2x+1)^2\right)^2}\,dx$$

$$= \frac{(2x+1)^2 - \left[4x(2x+1)\right]}{(2x+1)^4}$$

$$= \frac{(2x+1)\left((2x+1)-[4x]\right)}{(2x+1)^4}$$

$$= \frac{\left((2x+1)-4x\right)}{(2x+1)^3} \Rightarrow \frac{1-2x}{(2x+1)^3}\,dx$$

$$dv = e^{2x}dx$$

$$v = \frac{1}{2}e^{2x}$$

$$\int udv = uv - \int vdu$$

$$xe^{2x}\left(\frac{1}{2(2x+1)}\right) - \int -\frac{1}{2(2x+1)}\cdot e^{2x}(1+2x)\,dx$$

$$-\frac{xe^{2x}}{2(2x+1)} + \frac{1}{2}\int \frac{e^{2x}(1+2x)\,dx}{(2x+1)}$$

$$-\frac{xe^{2x}}{2(2x+1)} + \frac{1}{2}\int \frac{e^{2x}\cancel{(1+2x)}\,dx}{\cancel{(2x+1)}}$$

$$-\frac{xe^{2x}}{2(2x+1)} + \frac{1}{2}\int e^{2x}dx$$

$$\left[-\frac{xe^{2x}}{2(2x+1)} + \frac{1}{2}\cdot\frac{1}{2}e^{2x}\right]$$

$$-\frac{1}{2}e^{2x}\left[\frac{x}{(2x+1)} - \frac{1}{2}\right]$$

$$-\frac{1}{2}e^{2x}\left[\frac{2x-(2x+1)}{2(2x+1)}\right] + C$$

$$-\frac{1}{2}e^{2x}\left[\frac{-1}{2(2x+1)}\right] + C$$

$$\frac{e^{2x}}{4(2x+1)} + C$$

8.2#16

$$11\int \arctan x\,dx$$

$u = \arctan x \qquad dv = dx$

$\tan u = x \qquad \text{Triangle}\ \boxed{\begin{matrix} opp = x, adj = 1 \\ hyp = \sqrt{1+x^2}\end{matrix}}$

$du = \sec^2 u u' = 1$

$$u' = \frac{1}{\sec^2 u} = \frac{1}{\left(\frac{\sqrt{1+x^2}}{1}\right)^2} = \frac{1}{1+x^2}$$

$v = x$

$$\int u\,dv = uv - \int v\,du$$

$$\arctan x \cdot x - \int x \cdot \frac{1}{1+x^2}dx$$

$$x\arctan x - \int \frac{x}{1+x^2}dx$$

$$11\left[x\arctan x - \frac{1}{2}\ln\left|1+x^2\right| + C\right]$$

8.2#25

$$\int \cos(\ln x)\,dx$$

$u = \ln x\ du = \frac{1}{x}dx$

$du = \frac{1}{x}dx \qquad xdu = dx$

$u = \ln x \Rightarrow \qquad e^u du = dx$

$$\int \cos u e^u du$$

Proceed with integration by parts $\cdots$

Trigonometric Integrals

Random class example

$\int \sec^3 x \tan x dx$ $\boxed{\begin{aligned} u &= \sec x \\ du &= \sec x \tan x dx \end{aligned}}$ $\int u^2 du = \frac{1}{3}u^3 + C$ $= \frac{1}{3}\sec^3 x + C$ $\int \tan^2 x \sec^2 x dx$ $\boxed{\begin{aligned} u &= \tan x \\ du &= \sec x \tan x \end{aligned}}$ $\int u^2 du$ $\ldots$		

Random class example

$\int \sec^5 x \tan x dx$ $\int \sec^4 x \sec x \tan x dx$ $\boxed{\begin{aligned} u &= \sec x \\ du &= \sec x \tan x \end{aligned}}$ $\ldots$		

Random Class example.

$\int \frac{\sec^2 x}{5+7\tan^2 x}dx$ $\boxed{\begin{array}{l} a^2=5 \quad u^2=7\tan^2 x \\ u=\sqrt{7}\tan x \\ du=\sqrt{7}\sec^2 x \end{array}}$ $\boxed{\int \frac{du}{a^2+u^2}}$ $\frac{1}{\sqrt{7}}\cdot\frac{1}{\sqrt{5}}\arctan\frac{\left(\sqrt{7}\tan x\right)}{\sqrt{5}}$ $\frac{1}{\sqrt{35}}\arctan\frac{\left(\sqrt{7}\tan x\right)}{\sqrt{5}}+C$		

Integral involving sine-cosine products with different angles

Random example

$\int \sin 2x\cos 4x dx$ $\boxed{\begin{array}{l} \sin\alpha\cos\beta \Rightarrow \\ \sin(a+b)=\sin a\cos b+\sin b\cos a \\ \sin(a-b)=\sin a\cos b-\sin b\cos a \\ \hline\hline \sin(a+b)+\sin(a-b)=2\sin a\cos b \\ \frac{\sin(a+b)+\sin(a-b)}{2}=\sin a\cos b \end{array}}$ $\therefore \sin 2x\cos 4x=\frac{1}{2}\left(\sin 6x+\sin(-2x)\right)$ $\int \sin 2x\cos 4x dx \Rightarrow \frac{1}{2}\int\left(\sin 6x+\sin(-2x)\right)dx$ $\frac{1}{2}\int \sin 6x dx-\frac{1}{2}\int \sin 2x dx$		

8.3#20

$\int \sin(x)\cos(x)\,dx$ $\boxed{\begin{array}{l} u=\sin x \\ du=\cos x\,dx \end{array}}$ $\int u\,du=\frac{1}{2}u^2+C$ $=\frac{1}{2}\sin^2(x)+C$ $\boxed{\begin{array}{l} u=\cos x \\ du=-\sin x\,dx \end{array}}$ $\int -(du)u$ $-\frac{1}{2}u^2+C$ $-\frac{1}{2}\cos^2 x+1$ $-\frac{1}{2}\left(1-\sin^2 x\right)+1$ $-\frac{1}{2}+\frac{1}{2}\sin^2 x+C$ $\frac{1}{2}\sin^2 x+C-\frac{1}{2}=\frac{1}{2}\sin^2 x+C$	$\int \sin x\cos x\,dx$ $\boxed{\begin{array}{ll} u=\sin x & dv=\cos x\,dx \\ du=\cos x\,dx & v=\int \cos x\,dx \end{array}}$ $\int u\,dv=uv-\int v\,du$ $\sin x\cdot\sin x-\int \sin x\cos x\,dx$ $\boxed{\begin{array}{ll} u=\sin x & du=\cos x \end{array}}$ $\sin x\cdot\sin x-\int u\,du$ $\sin x\cdot\sin x-\frac{1}{2}u^2+C$ $\sin^2 x-\frac{1}{2}\sin^2 x+C$ $\frac{1}{2}\sin^2 x+C$	…

8.2#18

Find the integral $\int e^{3x}\cos 2x dx$ LIATE $\boxed{\begin{array}{ll} u=\cos 2x & dv=e^{3x}dx \\ du=-2\sin 2x dx & v=\frac{1}{3}e^{3x} \end{array}}$ $\int u dv = uv - \int v du$ $\cos 2x\cdot\frac{1}{3}e^{3x}-\int\frac{1}{3}e^{3x}\left(-2\sin 2x dx\right)$ $\frac{1}{3}e^{3x}\cos 2x+\frac{2}{3}\int e^{3x}\sin 2x dx$ $\boxed{\begin{array}{ll} u=\sin 2x & du=2\cos 2x dx \\ dv=e^{3x}dx & v=\frac{1}{3}e^{3x} \end{array}}$	$\frac{1}{3}e^{3x}\cos 2x+\frac{2}{3}\left[\sin 2x\cdot\frac{1}{3}e^{3x}-\int\frac{1}{3}e^{3x}2\cos 2x dx\right]$ $\int e^{3x}\cos 2x dx=\frac{1}{3}e^{3x}\cos 2x+\frac{2}{9}e^{3x}\sin 2x-\frac{4}{9}\int e^{3x}\cos 2x dx$ $\frac{13}{9}\int e^{3x}\cos 2x dx=\frac{1}{3}e^{3x}\cos 2x+\frac{2}{9}e^{3x}\sin 2x+C$ $\frac{9}{13}\left[\frac{13}{9}\int e^{3x}\cos 2x dx=\frac{1}{3}e^{3x}\cos 2x+\frac{2}{9}e^{3x}\sin 2x+C\right]$ $\int e^{3x}\cos 2x dx=\frac{3}{13}e^{3x}\cos 2x+\frac{2}{13}e^{3x}\sin 2x+C$ $\int e^{3x}\cos 2x dx=\frac{1}{13}e^{3x}\left[3\cos 2x+2\sin 2x\right]+C$	

Trigonometric Integrals.

Random Example

$\int \sin xdx = -\cos x + C$ $\int \sin^2 xdx =$ $\boxed{\begin{aligned} &\cos(A+B) = \cos A\cos B = \sin A\sin B \\ &\cos(A+A) = \cos A\cos A = \sin A\sin A \\ &\cos 2A = \cos^2 A - \sin^2 A \\ &\qquad = 1 - 2\sin^2 A \\ &\frac{-1+\cos 2A}{-2} = \frac{-2\sin^2 A}{-2} \\ &\frac{1}{2}\int (1-\cos 2x)dx = \frac{1}{2}\int dx - \frac{1}{2}\int \cos 2xdx \\ &\qquad = \frac{1}{2}x - \frac{1}{4}\int \cos udu \\ &\qquad = \frac{1}{2}x - \frac{1}{4}\sin 2x + C \end{aligned}}$	$\int \sin^3 xdx$ $\int \sin^2 x\sin xdx \Rightarrow \int u^n du$ $\int (1-\cos^2 x)\sin xdx$ $\int \sin xdx - \int \cos^2 x\sin xdx$ $\boxed{\begin{aligned} &u = \cos x \\ &du = -\sin xdx \end{aligned}}$ $\int \sin xdx - \int u^2 du$ $-\cos xdx - \frac{1}{3}u^3 + C$ $-\cos xdx - \frac{1}{3}(\cos x)^3 + C$ $\int (\sin^2 x)^2 dx$ $\int \left(\frac{1-\cos 2x}{2}\right)^2 dx$ $\int \frac{1-2\cos 2x+\cos^2 2x}{4}dx$ $\frac{1}{4}\int dx - \frac{1}{4}\cdot 2\int \cos 2x + \frac{1}{4}\int \cos^2 2xdx$ $\frac{1}{4}x - \frac{1}{4}\sin 2x + \frac{1}{4}\int \left(\frac{1+\cos 4x}{2}\right)dx$ $\frac{1}{4}x - \frac{1}{4}\sin 2x + \frac{1}{4}\cdot\frac{1}{2}\int dx + \frac{1}{4}\cdot\frac{1}{4}\cdot\frac{1}{2}\int \cos 4x4dx$ $\frac{3}{8}x - \frac{1}{4}\sin 2x + \frac{1}{32}\sin 4x + C$	

Integrals Involving powers of secant and tangent.

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\int \tan x \sec^2 x dx$$

$$\boxed{\begin{array}{l} u = \tan x \\ du = \sec^2 x \end{array}} \; or \; \boxed{\begin{array}{l} u = \sec x \\ du = \sec x \tan x \end{array}}$$

$$\int u du = \frac{1}{2}u^2 + C$$

$$= \frac{1}{2}\sec^2 x + C$$

$$= \frac{1}{2}(\tan^2 x + 1) + C$$

$$= \frac{1}{2}\tan^2 x + C$$

$$\int \sec x dx = \ln|\sec x + \tan x| + C$$

$$\int \sec^2 x = \tan x + C$$

$$\int \sec^3 x = \int \sec^2 x \sec x dx$$

$$\boxed{\begin{array}{ll} u = \sec x & dv = \sec^2 x dx \\ du = \sec x \tan x dx & \\ v = \tan x & \end{array}}$$

$$\int \sec^3 x = \sec x \tan x - \int \tan x \sec x \tan x dx$$

$$\int \sec^3 x = \sec x \tan x - \int \tan^2 x \sec x dx$$

$$\int \sec^3 x = \sec x \tan x - \int (\sec^2 x - 1)\sec x dx$$

$$\int \sec^3 x = \sec x \tan x - \int \sec^3 x dx + \int \sec x dx$$

$$2\int \sec^2 x = \sec x \tan x - \int \sec^3 x dx$$

$$= \frac{1}{2}\left[\int \sec^2 x = \sec x \tan x - \ln|\sec x + \tan x|\right]$$

$$\int \sec^2 x = \frac{1}{2}\sec x \tan x - \frac{1}{2}\ln|\sec x + \tan x|$$

Monday, March 4, 2024

$\int \frac{x}{x^2-5}dx$ $\boxed{\begin{array}{l} x=\sqrt{5}\sec\theta \\ dx=\sqrt{5}\sec\theta\tan\theta d\theta \\ x^2-5=\left(\sqrt{5}\sec\theta\right)^2-5 \\ =5\left(\sec^2\theta-1\right) \\ but\ \sec^2\theta-1=\tan^2\theta \\ =5\tan^2\theta \end{array}}$ $\int \frac{x}{x^2-5}dx=\int \frac{\sqrt{5}\sec\theta}{5\tan^2\theta}\left(\sqrt{5}\sec\theta\tan\theta d\theta\right)$ $\int \frac{\sec\theta\sec\theta}{\tan\theta}d\theta$ $\int \frac{\left(\tan^2\theta\right)+1}{\tan\theta}d\theta$ $\int\left(\tan\theta+\cot\theta\right)d\theta$ $=\ln\left\|\sec\theta\right\|+\ln\left\|\sin\theta\right\|+C$	$=\ln\left\|\sec\theta\right\|+\ln\left\|\sin\theta\right\|+C$ $\boxed{\text{Triange:hyp=x,andj=}\sqrt{5},opp=\sqrt{x^2-5}}$ $\ln\left\|\frac{x}{\sqrt{5}}\right\|+\ln\left\|\frac{\sqrt{x^2-5}}{x}\right\|+C$ $\ln\left\|\frac{x}{\sqrt{5}}\cdot\frac{\sqrt{x^2-5}}{x}\right\|+C$ $\ln\left\|\frac{\sqrt{x^2-5}}{\sqrt{5}}\right\|+C=\ln\left\|\sqrt{x^2-5}\right\|-\ln\sqrt{5}+C$ $\ln\left\|\sqrt{x^2-5}\right\|+C$	

Same Problem but using U-sub

$\int \frac{x}{x^2-5}dx$ $\boxed{\begin{array}{l} u=x^2-5 \\ du=2x \end{array}}$ $\frac{1}{2}\int\frac{2xdx}{x^2-5}$ $\frac{1}{2}\int\frac{du}{u}=\frac{1}{2}\ln\left\|u\right\|+C$ $\frac{1}{2}\ln\left\|x^2-5\right\|+C$ $\ln\sqrt{x^2-5}+C$		

Random example

<table>
<tr>
<td>
$3\int \sec^2\theta \cdot \sec\theta d\theta$

$\boxed{\begin{array}{ll} u=\sec\theta & dv=\sec^2\theta d\theta \\ v=\tan\theta & du=\sec\theta\tan\theta d\theta \end{array}}$

$3\left[\sec\theta\tan\theta-\int\tan\theta\sec\theta\tan\theta d\theta\right]$

$3\int\sec^3\theta d\theta=3\left[\sec\theta\tan\theta-\int\tan\theta\sec\theta\tan\theta d\theta\right]$

$=3\left[\sec\theta\tan\theta-\int\tan^2\theta\sec\theta d\theta\right]$

$=3\left[\sec\theta\tan\theta-\int\left(\sec^2\theta-1\right)\sec\theta d\theta\right]$

$=3\left[\sec\theta\tan\theta-\int\sec^3\theta-\sec\theta d\theta\right]$

$6\int\sec^3\theta d\theta=3\sec\theta\tan\theta+3\ln\left|\sec\theta+\tan\theta\right|+C$

$=\frac{1}{2}\sec\theta\tan\theta+\frac{1}{2}\ln t\left|\sec\theta+\tan\theta\right|+C$
</td>
<td>
$\boxed{\begin{array}{l} \text{triangle:opp=x,adj=}\sqrt{3}, \\ hyp=\sqrt{3+x^2} \end{array}}$

$=\frac{1}{2}\sec\theta\tan\theta+\frac{1}{2}\ln t\left|\sec\theta+\tan\theta\right|+C=$

$\frac{1}{2}\cdot\frac{\sqrt{3+x^2}}{\sqrt{3}}\cdot\frac{x}{\sqrt{3}}+\frac{1}{2}\ln\left|\frac{\sqrt{3+x^2}}{\sqrt{3}}+\frac{x}{\sqrt{3}}\right|+C$

$=\frac{1}{6}\cdot x\sqrt{3+x^2}+\frac{1}{2}\ln\left|\frac{x+\sqrt{3+x^2}}{\sqrt{3}}\right|+C$

$=\frac{1}{2}\ln\left|x+\sqrt{3+x^2}\right|-\frac{1}{2}\ln\sqrt{3}+C$

$=\frac{1}{6}x\sqrt{3+x^2}+\frac{1}{2}\ln\left|x+\sqrt{3+x^2}\right|+C$
</td>
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</table>

8.4#17

Evaluate the integral using the given integration limits. $\int_3^6 \frac{\sqrt{x^2-6}}{x^2}dx$ $x = 3\sec\theta \qquad dx = 3\sec\theta\tan\theta d\theta$ $x^2 = 9\sec^2\theta$ $\sqrt{x^2-9} = \sqrt{9\sec^2\theta - 9}$ $\sqrt{x^2-9} = \sqrt{9\left(\sec^2\theta - 1\right)}$ $= 3\sqrt{\tan^2\theta} = 3\tan\theta$ $if x = 3, \Rightarrow 3\sec\theta = 3$ $\theta = 0$ If x=6, then 6=3secθ 2=secθ $\frac{\pi}{3} = \theta$ $\int_0^{\pi/3} \frac{3\tan\theta}{9\sec^2\theta} \cdot \left(3\sec\theta\tan\theta d\theta\right)$ $\int_0^{\pi/3} \frac{\tan^2\theta}{\sec\theta} d\theta$ $\int_0^{\pi/3} \frac{\sec^2\theta - 1}{\sec\theta} d\theta$ $\int_0^{\pi/3} \left(\sec\theta - \cos\theta\right) d\theta$ $\int_0^{\pi/3} \left[\ln\left\|\sec\theta + \tan\theta\right\| - \sin\theta\right] d\theta$	$= \ln\left\|2+\sqrt{3}\right\| - \frac{\sqrt{3}}{2} - \left(\ln\left\|1+0\right\| - 0\right)$ $= \ln\left\|2+\sqrt{3}\right\| - \frac{\sqrt{3}}{2}$	

Tuesday, March 5, 2024

8.2#11

$\int \frac{x^3 e^{x^2}}{\left(x^2+1\right)^2} dx$ $\int \frac{x}{\left(x^2+1\right)^2} \cdot x^2 e^{x^2} dx$ $\boxed{\begin{array}{ll} u = x^2 e^{x^2} & dv = \frac{x}{\left(x^2+1\right)^2} dx \\ du = 2x \cdot e^{x^2} + e^{x^2} \cdot 2x \cdot x^2 & \\ = 2xe^{x^2}\left(1+x^2\right) & \\ v = \frac{1}{2}\int \frac{2xdx}{\left(x^2+1\right)^2} & \end{array}}$ $= uv - \int vdu$ $= x^2 e^{x^2} - \frac{1}{2\left(x^2+1\right)} - \int -\frac{1}{2\left(x^2+1\right)} \ldots$ $= -\frac{1}{2} \cdot \frac{x^2 e^{x^2}}{x^2+1} + \int xe^{x^2} dx$ $\boxed{\begin{array}{l} u = x^2 \\ du = 2xdx \end{array}}$ $= -\frac{x^2 e^{x^2}}{2\left(x^2+1\right)} + \frac{1}{2}\int e^u du$	$= -\frac{x^2 e^{x^2}}{2\left(x^2+1\right)} + \frac{1}{2} e^{x^2} + C$ $\ldots$ $= \frac{e^{x^2}}{2\left(x^2+1\right)} + C$ CHECK WITH DERIVATION. $\frac{d}{dx}\left[\frac{e^{x^2}}{2\left(x^2+1\right)}\right] = \frac{1}{2} \cdot \left[\frac{2xe^{x^2}\left(x^2+1\right) - \left(2x \cdot e^{x^2}\right)}{\left(x^2+1\right)^2}\right]$ $\frac{1}{2} \cdot \left[\frac{2xe^{x^2}\left[x^2+1-1\right]}{\left(x^2+1\right)^2}\right]$ $\left[\frac{xe^{x^2} \cdot x^2}{\left(x^2+1\right)^2}\right] = \frac{x^3 e^{x^2}}{\left(x^2+1\right)^2}$ CONFIRMED	

8.2 #25

$\int \cos\left(\ln x\right) dx$ $\boxed{\begin{array}{ll} u = \ln x \Rightarrow x = e^u & \\ du = \frac{1}{xdx} & xdu = dx \\ x = e^u & e^u du = dx \end{array}}$ $\int \cos u \cdot e^u du$ $\int e^u \cos u du$ $\ldots$		

8.3#8

$\int \sin^{10}(x)$ $\int \left(\sin^2 x\right)^5 dx$ $\int \left[\frac{1-\cos 2x}{2}\right]^5 dx$ $\ldots$*very* tideous Wallis's Formular is easier way of managing the situation. $\int_0^{\pi/2} \sin^{10} x = \frac{1}{2}\cdot\frac{3}{4}\cdot\frac{5}{6}\cdot\frac{7}{8}\cdot\frac{9}{10}\cdot\frac{\pi}{2}$ $\frac{63\pi}{512}$		

Trigonometric Substitution.

Derivation of formular

$\int \frac{du}{\sqrt{u^2+a^2}}$ $\boxed{\begin{aligned}&u = a\tan\theta\\ &du = a\sec^2\theta d\theta\end{aligned}}$ $\sqrt{u^2+a^2} = \sqrt{(a\tan\theta)^2+a^2}$ $= \sqrt{a^2\tan^2\theta + a^2}$ $= \sqrt{a^2\left(\tan^2\theta+1\right)}$ $= \sqrt{a^2\sec^2\theta}$ $a\sec\theta$ $\int \frac{a\sec^2\theta d\theta}{a\sec\theta} = \int \sec\theta d\theta$ $\ln\lvert\sec\theta+\tan\theta\rvert + C$ Remember $\boxed{u = a\tan\theta}$ $\therefore \boxed{\text{triangle:opp=u,adj=a,hyp=}\sqrt{u^2+a^2}}$ $\ln\lvert\sec\theta+\tan\theta\rvert + C = \ln\left\lvert\frac{\sqrt{u^2+a^2}}{a} + \frac{u}{a}\right\rvert + C$ $\ln\left\lvert\frac{\sqrt{u^2+a^2}+u}{a}\right\rvert + C$ $\ln\left\lvert\sqrt{u^2+a^2}+u\right\rvert - \ln\lvert a\rvert + C$ $\ln\left\lvert\sqrt{u^2+a^2}+u\right\rvert + C$	$\therefore \boxed{\int \frac{du}{\sqrt{u^2+a^2}} = \ln\left\lvert\sqrt{u^2+a^2}+u\right\rvert + C}$	

Derivation of formular

$\int \frac{du}{a^2-u^2}$ $\boxed{\begin{array}{l} u = a\sin\theta \qquad du = a\cos\theta d\theta \\ a^2-u^2 = a^2-(a\sin\theta)^2 \\ = a^2(1-\sin^2\theta) = a^2\cos^2\theta \end{array}}$ $\int \frac{du}{a^2-u^2} = \int \frac{a\cos\theta d\theta}{a^2\cos^2\theta} = \int \frac{d\theta}{a\cos\theta}$ $= \frac{1}{a}\int \sec\theta = \frac{1}{a}\ln\lvert\sec\theta+\tan\theta\rvert + C$ Rem $\quad \text{Sin}\theta = \frac{u}{a}$ $\boxed{\begin{array}{l} \text{Triangle;Adj=}\sqrt{a^2-u^2}, \\ hyp = a, opp = u \end{array}}$ $= \frac{1}{a}\ln\left\lvert \frac{a}{\sqrt{a^2-u^2}} + \frac{u}{\sqrt{a^2+u^2}} \right\rvert + C$ $= \frac{1}{a}\ln\left\lvert \frac{a+u}{\sqrt{a^2+u^2}} \right\rvert + C$	simplification… $= \frac{1}{2a}\ln\left\lvert\frac{a+u}{a-u}\right\rvert + C$	

8.4#16

$\int \frac{x}{\sqrt{x^2-16x+48}}dx$ $x^2-16x+64+48-64$ $-16 \div 2 = -8$ $(-8)^2 = 64$ $(x-8)^2-16$ $\int \frac{x}{\sqrt{(x-8)^2-16}}dx$ $x-8 = 4\sec\theta$ $\boxed{\text{Triange:hyp=x-8,adj=4,opp=}\sqrt{(x-8)^2-16}}$ $\frac{1}{4}dx = \sec\theta\tan\theta d\theta$ $dx = 4\sec\theta\tan\theta d\theta$ $\sqrt{(x-8)^2-16} = \sqrt{(4\sec\theta)^2-16}$ $= \sqrt{16\sec^2\theta-16}$ $= \sqrt{16\tan^2\theta}$ $= 4\tan\theta$	$\int \frac{8+4\sec\theta}{4\tan\theta}\cdot(4\sec\theta\tan\theta d\theta)$ $\int(8+4\sec\theta)(\sec\theta d\theta)$ $\int\left(8\sec\theta+4\sec^2\theta\right)d\theta$ $8\int\sec\theta d\theta+4\int\sec^2\theta d$ $8\ln\|\sec\theta+\tan\theta\|+4\tan\theta+C$ $8\ln\left\|\frac{x-8}{4}+\frac{\sqrt{x^2-16x+48}}{4}\right\|+4\frac{\sqrt{x^2-16x+48}}{4}+C$ $8\ln\left\|x-8+\sqrt{x^2-16x+48}\right\|+\sqrt{x^2-16x+48}+C$	

Wednesday, March 6, 2024

8.5#11

$\int \frac{\sec x^2}{\tan^2 x + 11\tan x + 30} dx$ $u = \tan x \qquad du = \sec^2 x dx$ $\int \frac{du}{u^2 + 11u + 30} dx$ $\int \frac{du}{u^2 + 11u + 30} dx = \int \frac{du}{(u+5)(u+6)}$ $\frac{1}{(u+5)(u+6)} = \frac{A}{u+5} + \frac{B}{u+6}$ $1 = A(u+6) + B(u+5)$ $\boxed{\begin{array}{l} u = -6 \;\; 1 = -B \Rightarrow \boxed{B = -1} \\ u = -5 \Rightarrow \boxed{1 = A} \end{array}}$ $\int \frac{du}{u+5} - \int \frac{du}{u+6}$ $\ln\lvert u+5 \rvert - \ln\lvert u+6 \rvert$ $= \ln\left\lvert \frac{u+5}{u+6} \right\rvert + C$	$\ln\left\lvert \frac{5 + \tan x}{6 + \tan x} \right\rvert + C$	

8.5#9

$\int \frac{\sin(x)}{\cos(x)+(\cos(x))^2}dx$ $\boxed{u=\cos x \qquad du=-\sin x dx}$ $-\int \frac{-\sin(x)}{\cos(x)+(\cos(x))^2}dx$ $-\int \frac{du}{u+u^2} = -\int\frac{du}{u(1+u)} = -\left[\int\frac{Adu}{u}+\int\frac{Bdu}{1+u}\right]$ $\frac{1}{u(1+u)}=\frac{A}{u}+\frac{B}{1+u}$ $1=A(1+u)+Bu$ $\boxed{\begin{array}{l} if\ u=0 \Rightarrow 1=1 \\ if\ u=-A\Rightarrow 1=-B\end{array}}$ $-\int\frac{du}{u}-\int\frac{-1du}{1+u}$	$\int\frac{du}{1+u}-\int\frac{du}{u}$ $=\ln\lvert 1+u\rvert-\ln\lvert u\rvert+C$ $=\ln\left\lvert\frac{1+u}{u}\right\rvert+C$ $=\ln\left\lvert\frac{1}{u}+1\right\rvert+C$ $=\ln\lvert\sec x+1\rvert+C$	

8.5#5

$\int\frac{x^2+108x+108}{x^3-4x}dx$ $x^2-4=x(x^2-4)=x(x-2)(x+3)$ $\frac{x^2+108x+10}{x^3-4x}=\frac{x^2+108x+10}{x(x-2)(x+3)}$ $\left[\frac{x^2+108x+10}{x(x-2)(x+3)}=\frac{A}{x}+\frac{B}{x-2}+\frac{C}{x+2}\right]x(x-2)(x+3)$ $x^2+108x+108=A(x-2)(x+2)+Bx(x+2)+Cx(x-2)$ $\boxed{\begin{array}{l} if\ x=0\Rightarrow 108=-4A\Rightarrow A=-27 \\ if\ x=2\Rightarrow \boxed{4+216+108=8B}\Rightarrow B=41 \\ if\ x=-2\Rightarrow \boxed{4-216+108=8C}\Rightarrow C=-113\end{array}}$ $\int\frac{x^2+108x+108}{x^3-4x}dx=-27\int\frac{dx}{x}+41\int\frac{dx}{x-2}-13\int\frac{dx}{x+2}$ $=-27\ln\lvert x\rvert+41\ln\lvert x-2\rvert-13\ln\lvert x+2\rvert+C$	$=\ln\lvert(x-2)^{41}\rvert-\left(\ln\lvert x^{27}\rvert+\ln\lvert x+2\rvert^{13}\right)+C$ $=\ln(x-2)^{41}-\left(\ln x^{27}\cdot(x+2)^{13}\right)+C$ $=\ln\left\lvert\frac{(x-2)^{41}}{x^{27}(x+2)^{13}}\right\rvert+C$	

8.5#8

<table>
<tr>
<td>
$\int \frac{x}{16x^4-1}dx$
$16x^4-1=\left(4x^2-1\right)\left(4x^2+1\right)$
$=\left(2x-1\right)\left(2x+1\right)\left(4x^2+1\right)$
$\int \frac{x}{16x^4-1}dx=\int\frac{Adx}{2x-1}+\int\frac{Bdx}{2x+1}+\int\frac{Cx+Ddx}{4x^2+1}$
$\left[\frac{x}{16x^4-1}=\frac{A}{2x-1}+\frac{B}{2x+1}+\frac{Cx+D}{4x^2+1}\right]\left(16x^4-1\right)$
$x=A\left(2x+1\right)\left(4x^2+1\right)+B\left(2x-1\right)\left(4x^2+1\right)+\left(Cx+D\right)\left(4x^2-1\right)$
$if\ x=0\Rightarrow 0=A(1)+B(-1)+D(-1)$
$0=a-b-d \qquad (i)$
$if\ x=0\Rightarrow A(3)(5)+B(1)(5)+(C+D)3$
$1=15a+5b+3c+3d \qquad (iii)$
$if\ x=-\frac{1}{2}\Rightarrow -\frac{1}{2}=B(-2)(2)$
$b=\frac{1}{8}$
$if\ x=\frac{1}{2}\Rightarrow \frac{1}{2}=A(2)(2) \qquad \frac{1}{2}=4A$
From (i)
$0=\frac{1}{8}-\frac{1}{8}-d$
$d=0 \qquad (ii)$
From (iii) $\quad 1=15\cdot\frac{1}{8}+5\cdot\frac{1}{8}+3C$
$c=-\frac{1}{2}$
</td>
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$\int\frac{\frac{1}{8}dx}{2x-1}+\int\frac{\frac{1}{8}dx}{2x+1}+\int\frac{-\frac{1}{2}dx}{4x^2+1}$
$u=2x-1$
$du=2dx$
$=\frac{1}{2}\cdot\frac{1}{8}\int\frac{2dx}{2x-1}+\frac{1}{2}\cdot\frac{1}{8}\int\frac{2dx}{2x+1}-\frac{1}{2\cdot 8}\int\frac{8xdx}{4x^2-1}$
$=\frac{1}{16}\ln\left|\left(4x^2-1\right)\right|-\frac{1}{16}\ln\left(4x^2\right)$
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